Assignment No. 2

Q.1 Explain the characteristics of memory system and describe the memory hierarchy with diagram

Characteristics of memory system

① Capacity
- Amount of data memory can store
- Measured in bytes (KB, MB, GB, TB)

② Access Time
- Time required to read / write data
- Lower access time = faster memory

③ Volatility
- Volatile memory loses data when power is off (RAM)
- Non-volatile retains data (ROM, HDD)

④ Reliability
- Ability to store the data accurately without errors

Heirarchy of main memory

Memory hierarchy is the management of different types of memory in a computer system based on -
speed
cost
size (capacity)

Registers
↓
Cache memory

Main Memory

Secondary storage

Tertiary Storage.

fig. Hierarchy of memory.

① Register
- Located inside CPU
- Faster memory
- very small capacity
- Store immediate data for processing
ex. program counter

② Cache memory
- Between CPU and RAM
- stores frequently used data
- very high speed
- Types L1, L2, L3 cache

③ Main Memory (RAM)
- stores currently running programs
- Medium speed
- Volatile (data lost when power off)

(4) secondary storage
- Large capacity storage
- slower than RAM
- Non-volatile
ex. HDD, SSD

(5) Tertiary storage
- Backup storage
- very slow but huge capacity
- ex. magnetic tape, cloud backup

Advantages
- Improve performance
- Reduce cost
- Provides cost
- Provide large storage with fast access
- Efficient data management

Q.2 Difference between SRAM and DRAM

Aspects	SRAM	DRAM
(1) Full form	Static RAM	Dynamic RAM
(2) speed	faster	slower
(3) cost	Expensive	cheap
(4) Density	low	High

⑤ Storage	Flip Flops	Capacitors
⑥ Power	Less refresh needed	Needs refresh
⑦ Used in	Cache memory	Main memory

Q.3 Explain the types of secondary storage devices

① Magnetic storage Devices
- These devices store data using magnetic fields on a rotating surface
- large storage capacity
- low cost per GB
- ex. HDD, SSD, floppy disk

② Optical Storage Devices
- Optical storage uses laser light to read and write data
- Used for multimedia storage
- Moderate speed
- Resistant to magnetic interference
- ex. CD, DVD, Blue-ray Disc

③ Solid state Storage Devices
- stores data using electronic circuits (Flash memory) without moving parts
- ex. SSD, USB Flash drive
- very fast access speed

- Durable and shock resistant
- low power consumption
- More expensive than HDD

④ Cloud storage (Modern secondary storage)
- Data stored on remote servers and accessed through internet
- Accessible anywhere
- Scalable storage
- Requires internet connection

ex. google drive, one drive, Dropbox

Q.7. Compare different techniques to resolve cache memory

Aspects	Write through	Write Back
① Working	Data written to cache & main memory simultaneously	Data written only to cache, memory updated later
② Speed	slower (frequent memory access)	Faster (less memory access)
③ Memory Traffic	High	Low
④ Complexity	simple design	More complex

⑤ Data consistency	Memory always updated	Memory updated only when needed
⑥ Power Consumption	Higher	Lower
⑦ Usage	Simple system	High performance system

Q5. Explain RAID Architecture and describe levels 1 - 6 with example and advantages

RAID stands for Redundant Array of Independant Disks
It is a technology using combine multiple physical disks into a a single logical unit to improve
- Performance
- Data reliability
- storage capacity
- It distributes data access several disks using techniques like stripping, mirroring and parity

- RAID architecture organizes multiple disks and control how data is stored and retrieved.

Techniques Used

① Stripping - Data divided into blocks and stored across multiple disks, Improved speed

Mirroring
- Same data copied on multiple disks
- Improve reliability

Parity
- Extra information stored for error detection and recovery

RAID levels

① RAID 0 - Data split across disks
Advantages
- High performance
- Fast read/write speed

② RAID 1 (Mirroring) - Data duplicated on two disks
Advantages
- High reliability
- Easy recovery

③ RAID 2 - Bit level stripping with error correction
Advantages
- Uses hamming code
- Rarely used today

④ RAID 3 - Bite-level stripping, pedicated parity disk
Advantages
- Good for large sequential data transfer

(4) RAID 4 - Block level striping, single parity disk
Advantages
- Faster reads
- efficient raid level

(5) RAID 5 - Block level striping, distributed parity across all d
Advantages
- Balanced performance
- Fault tolerance
- Efficient storage use

(6) RAID 6 - similar to RAID 5 but with dual parity
Advantages
- Can survive two disk failures
- Higher reliability.